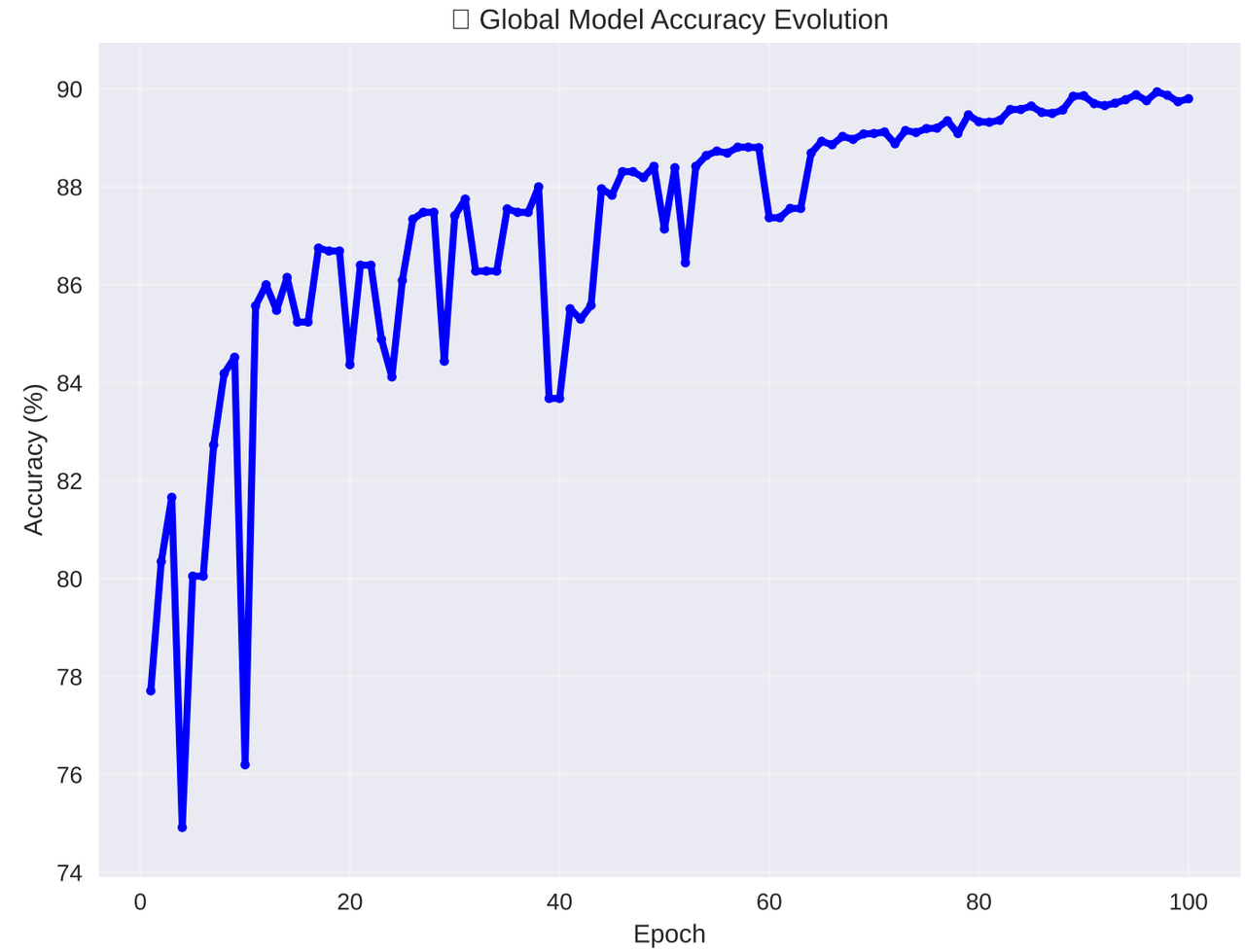
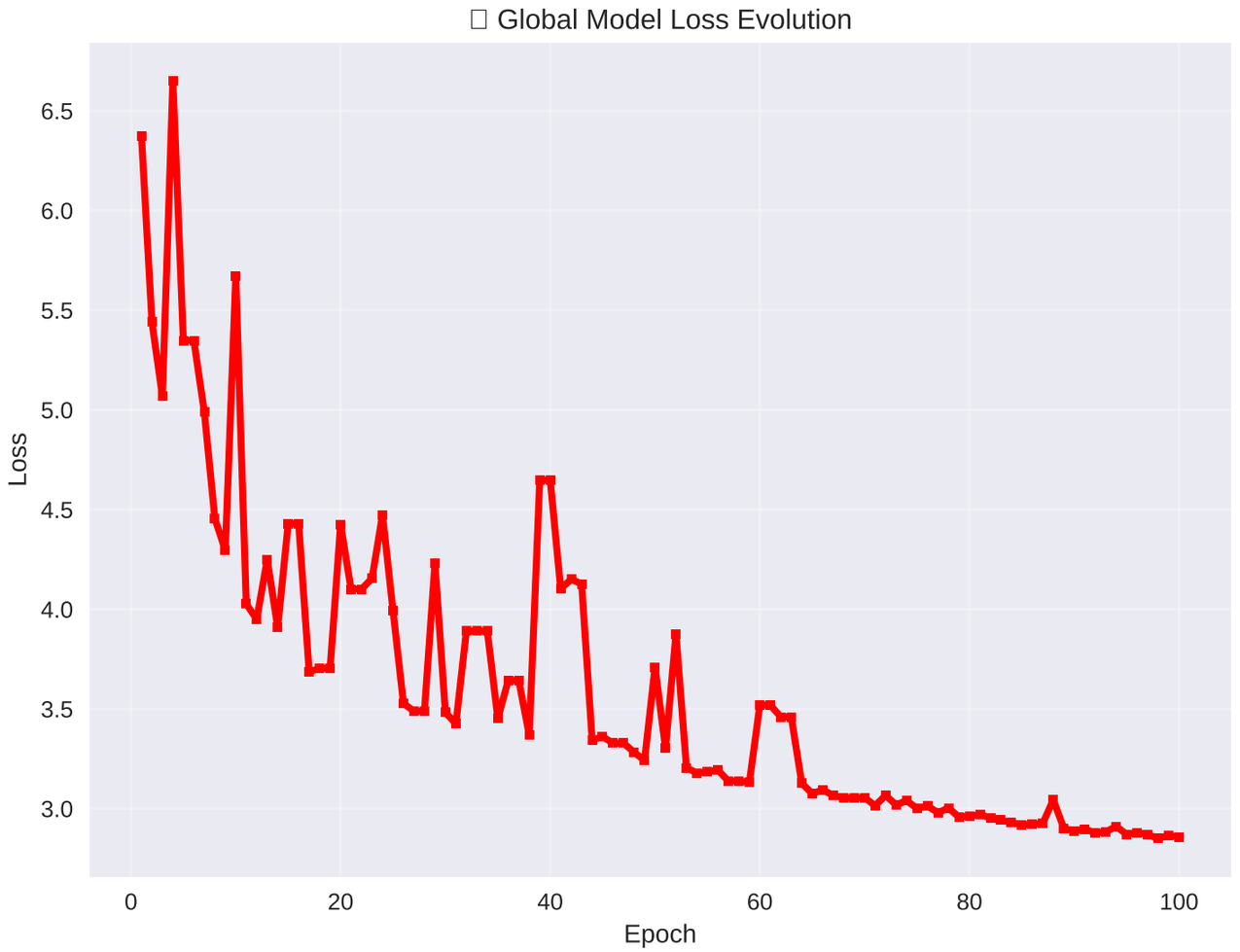
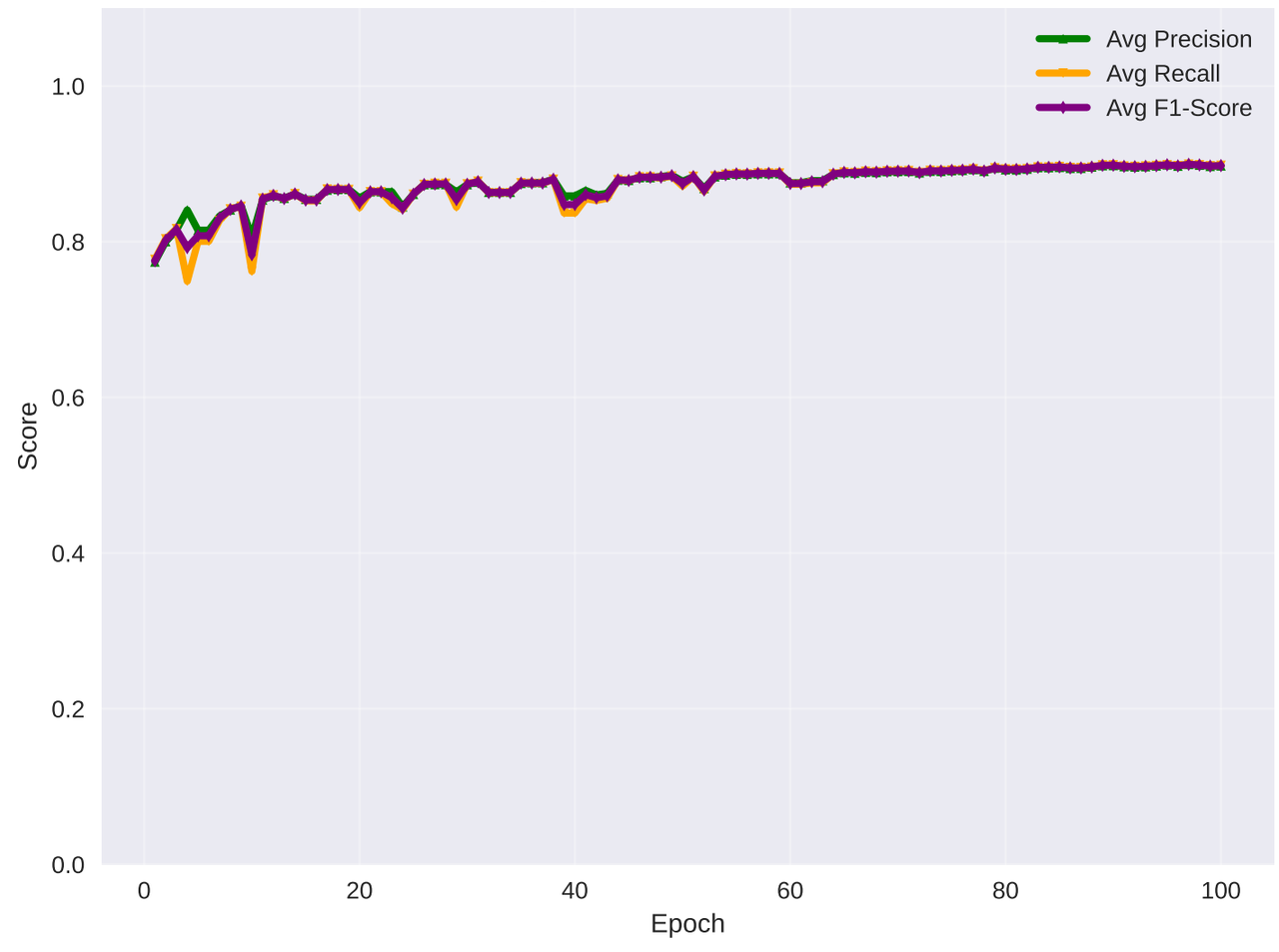
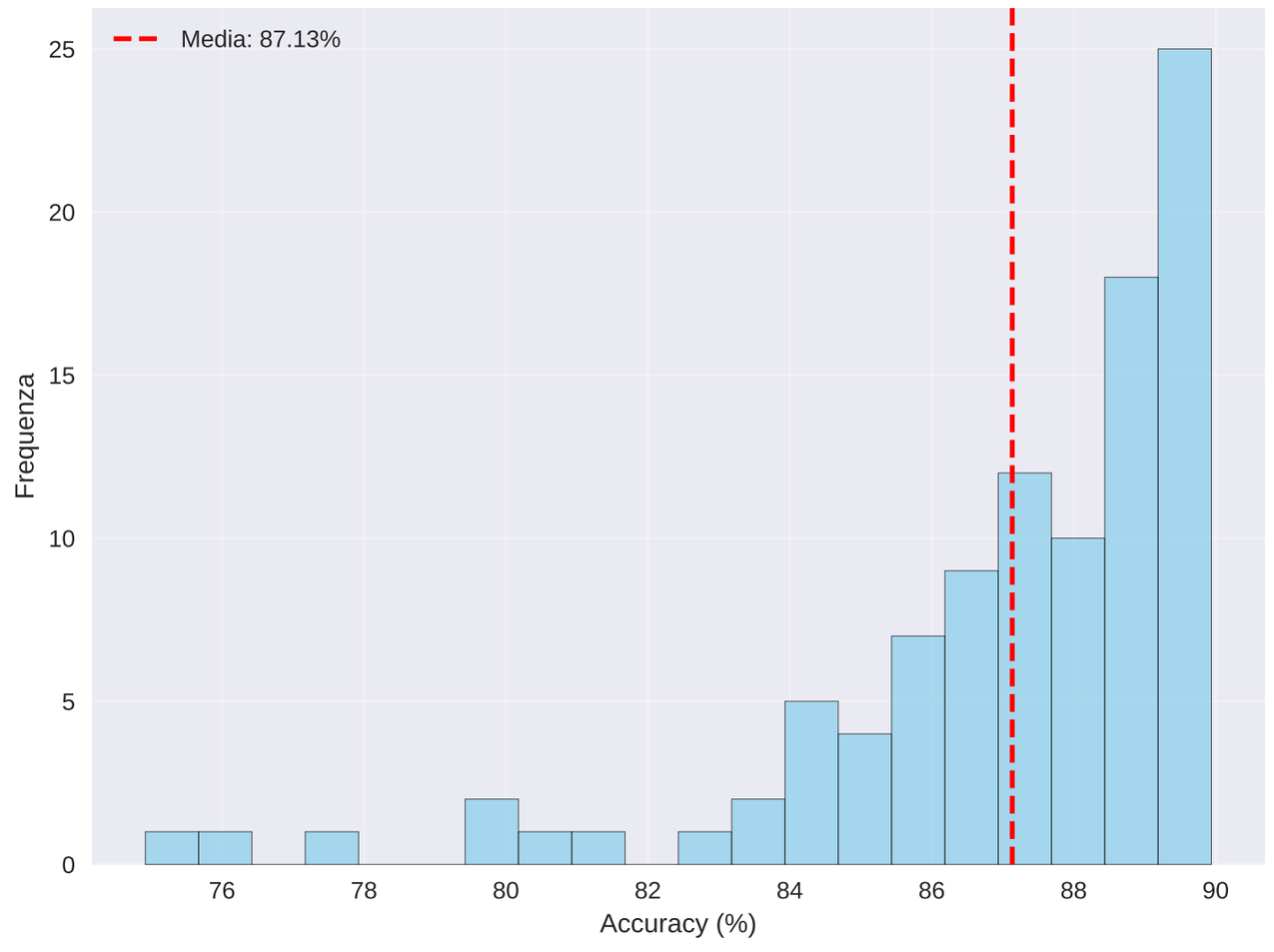
Federated Learning - Main Metrics Overview



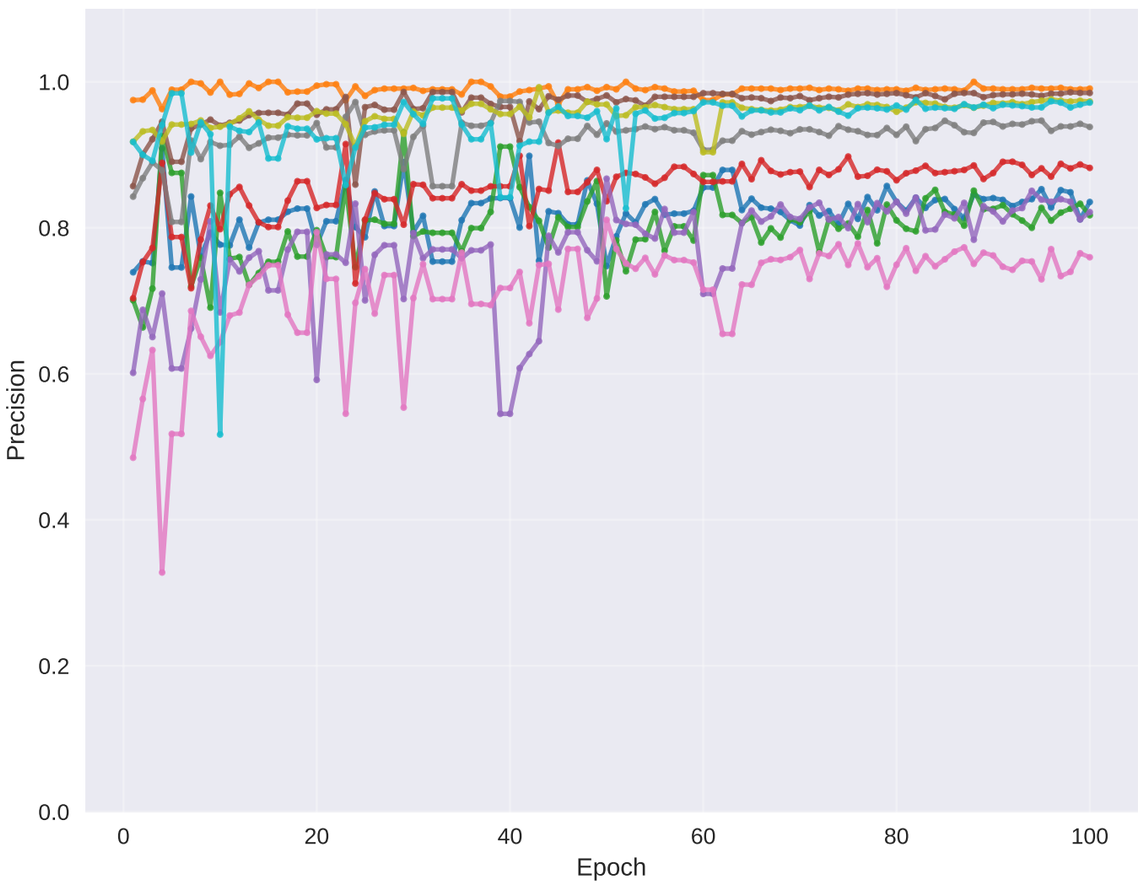




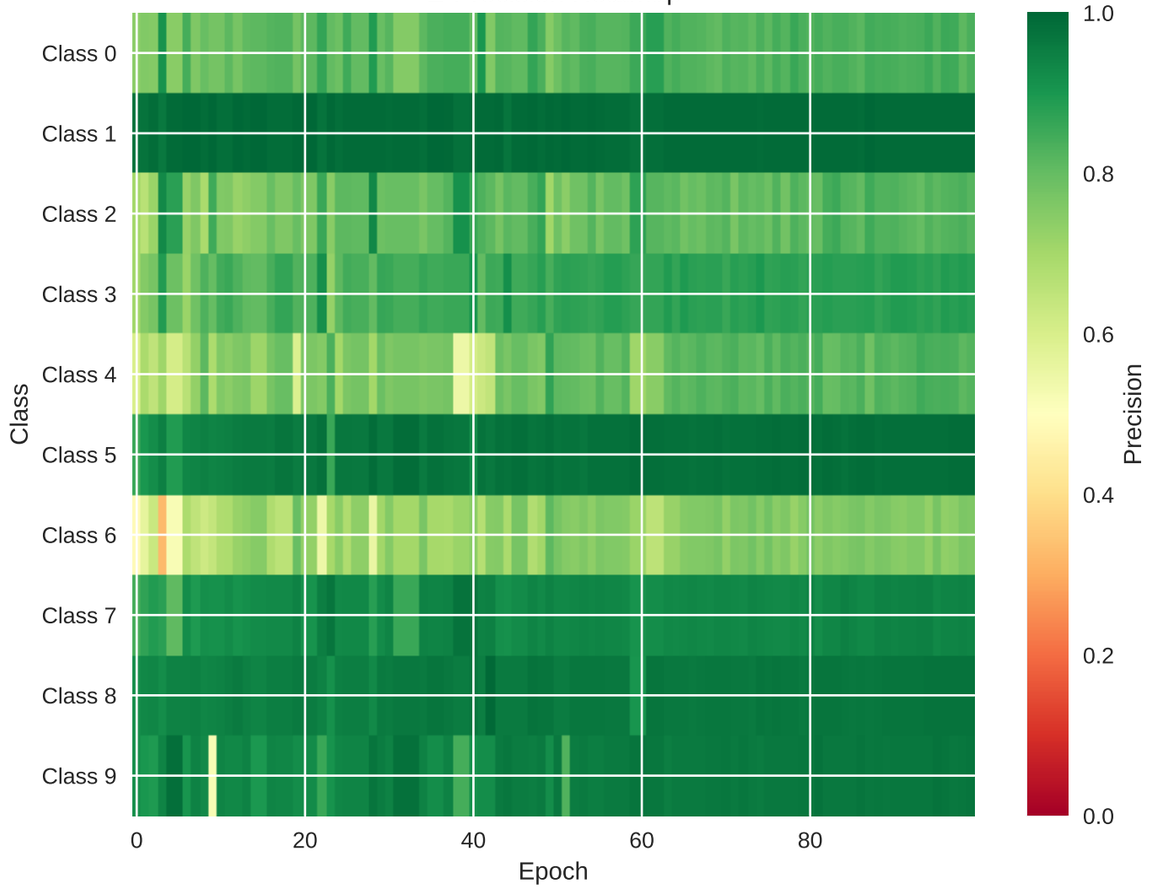


Per-Class Precision Analysis

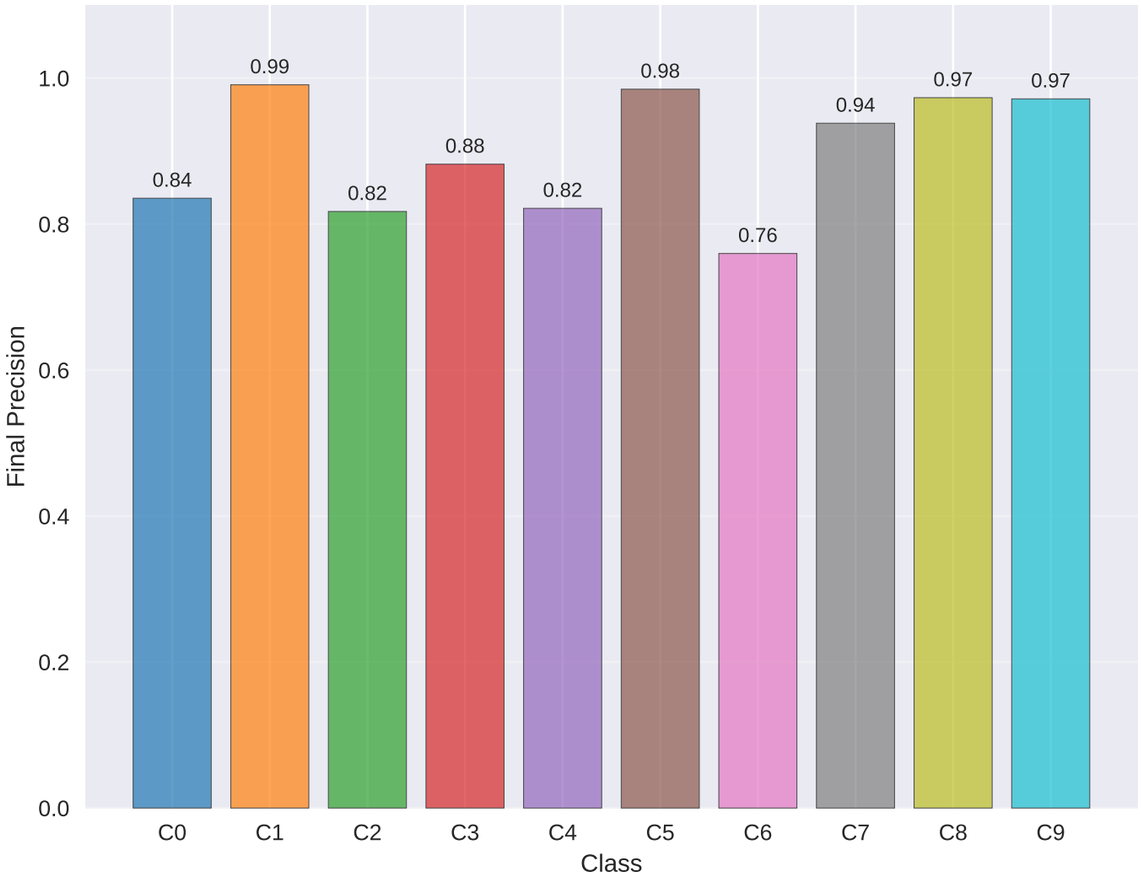
□ Per-Class Precision Evolution



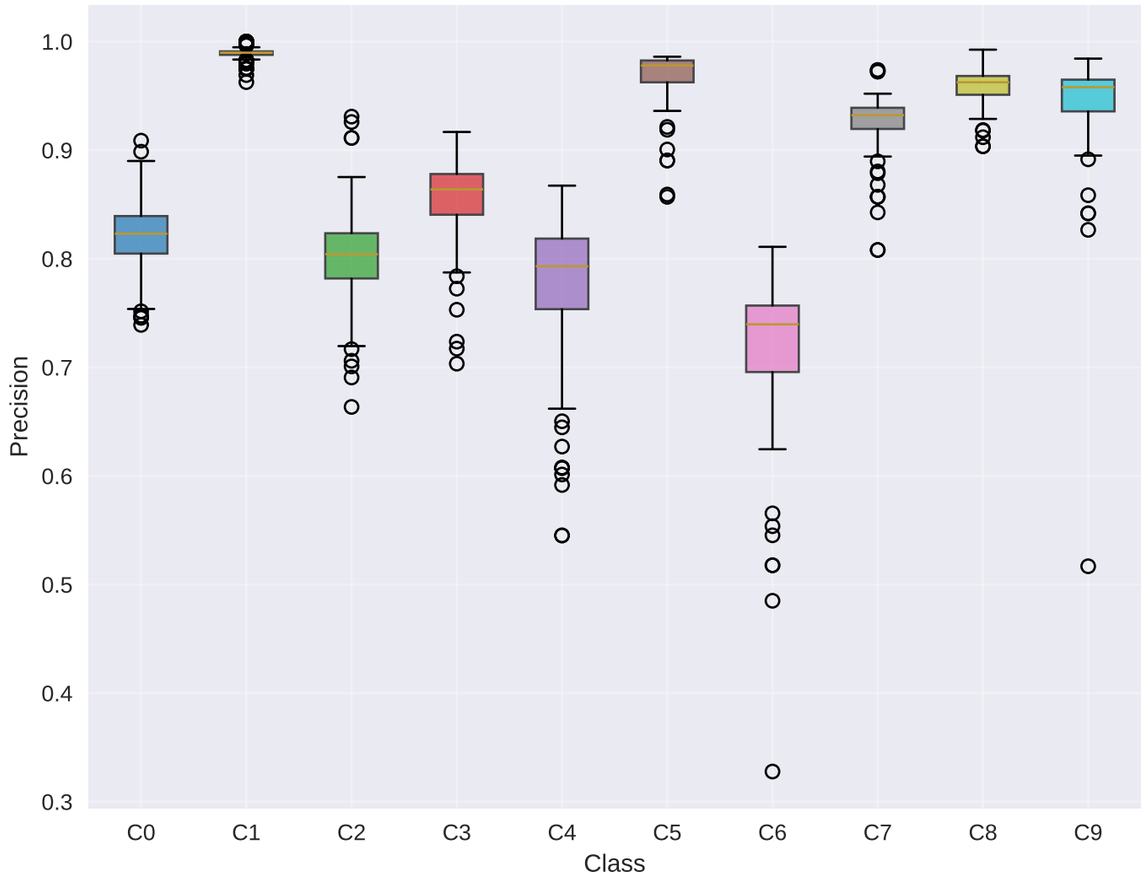
□ Precision Heatmap



□ Final Precision per Class

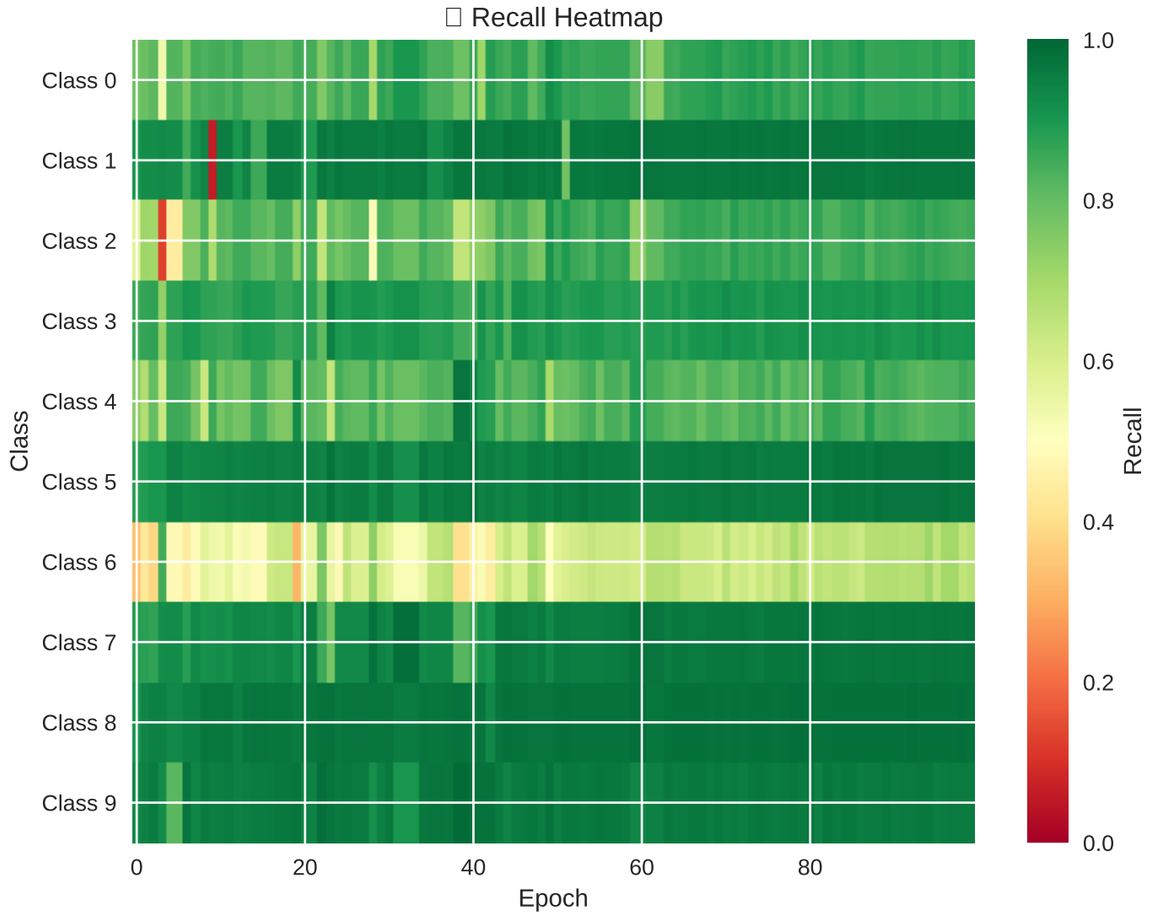


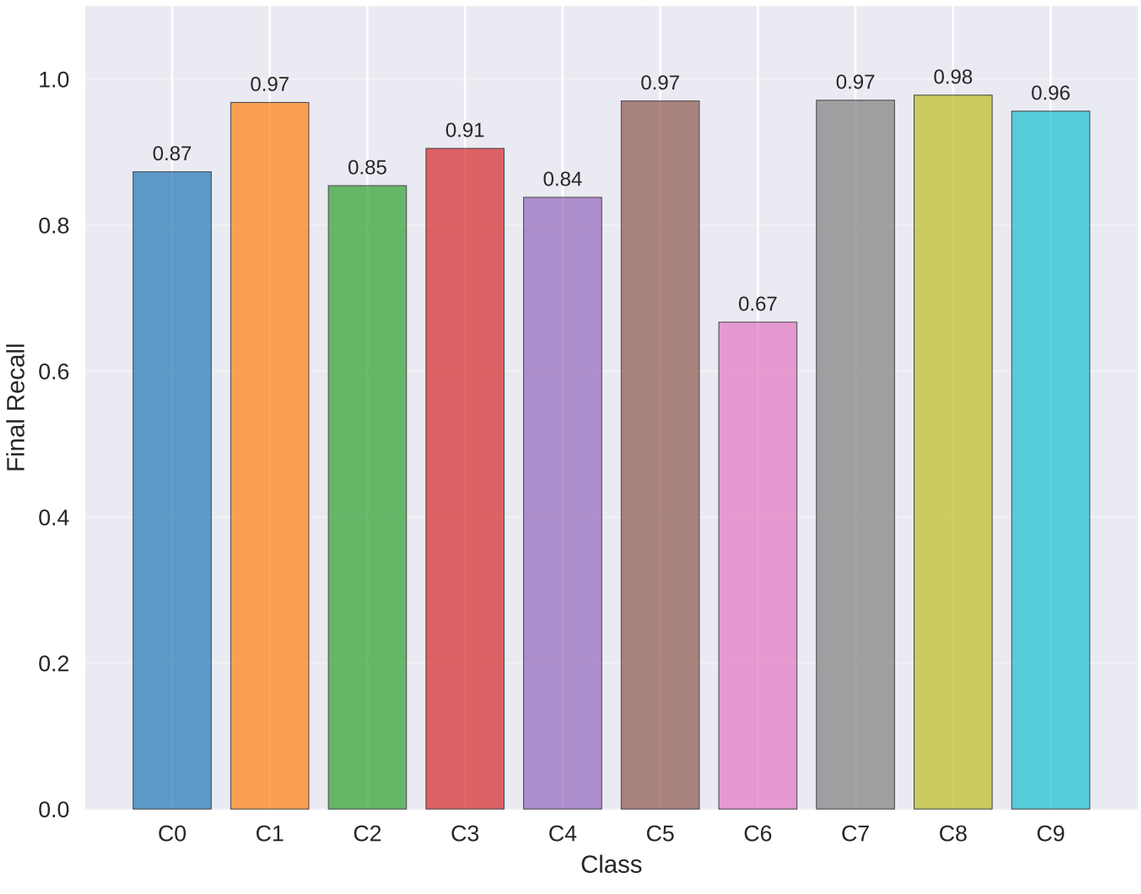
□ Precision Distribution per Class

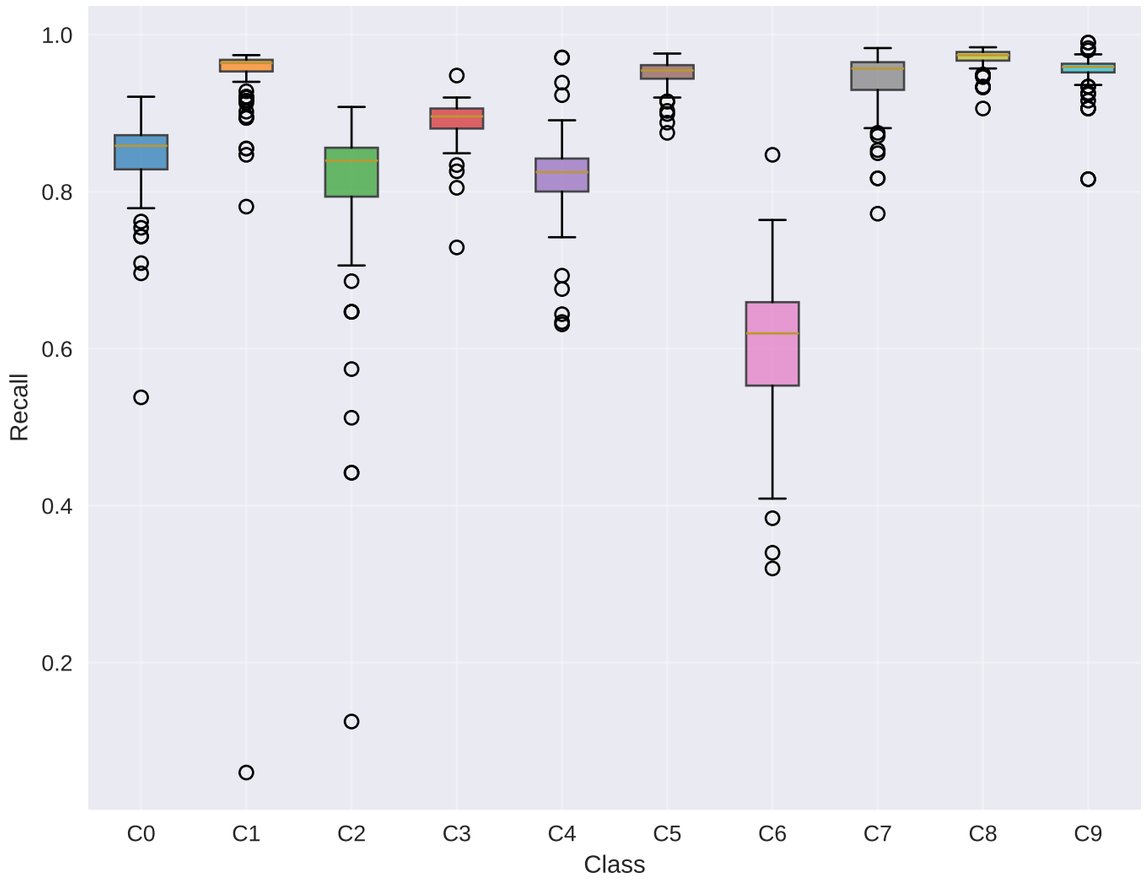


Per-Class Recall Analysis



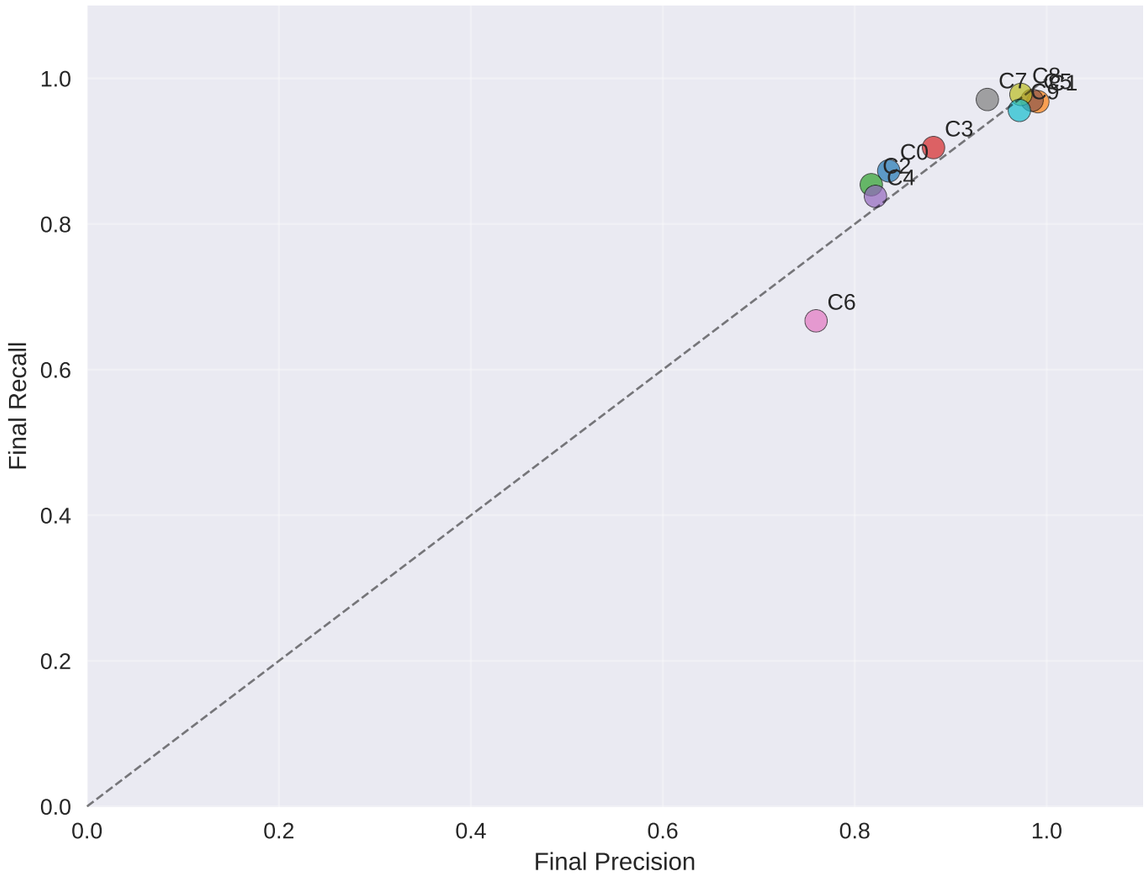






Comparative Analysis

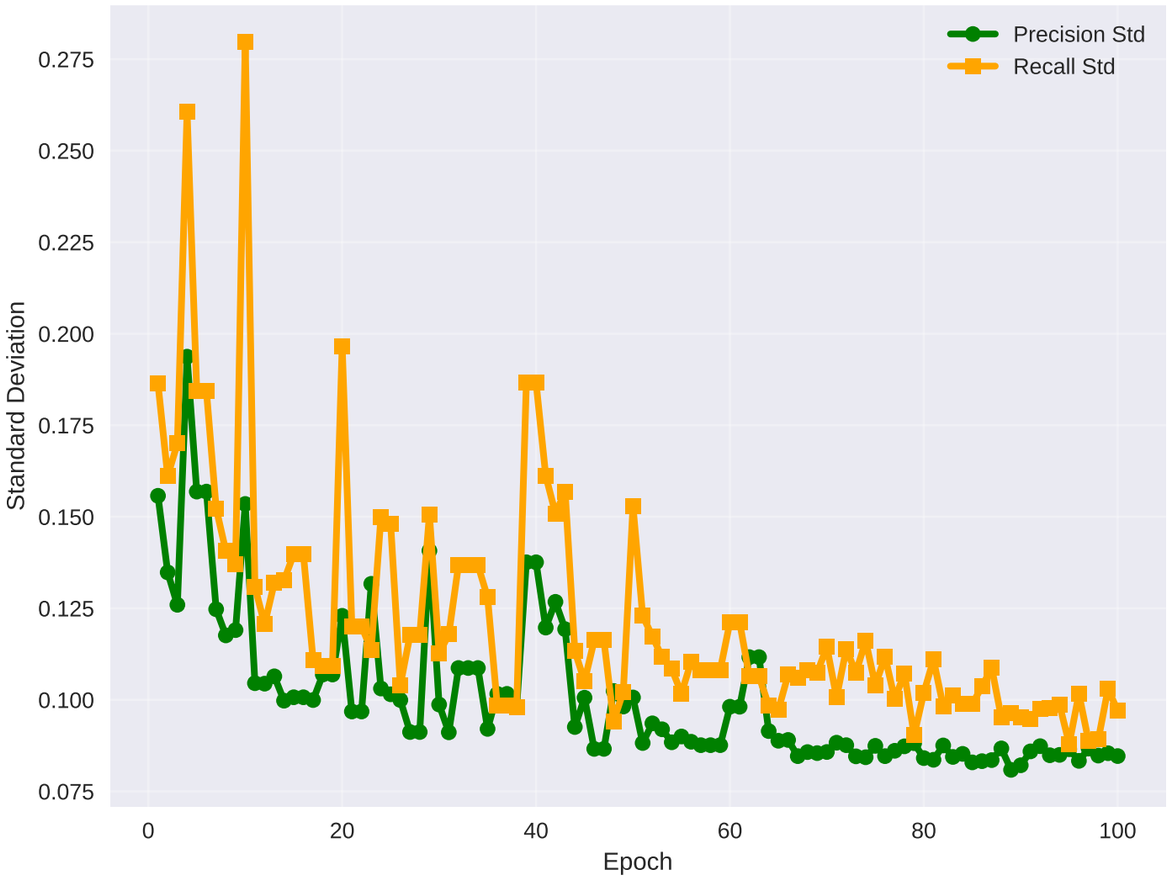
□ Precision vs Recall (Final Values)



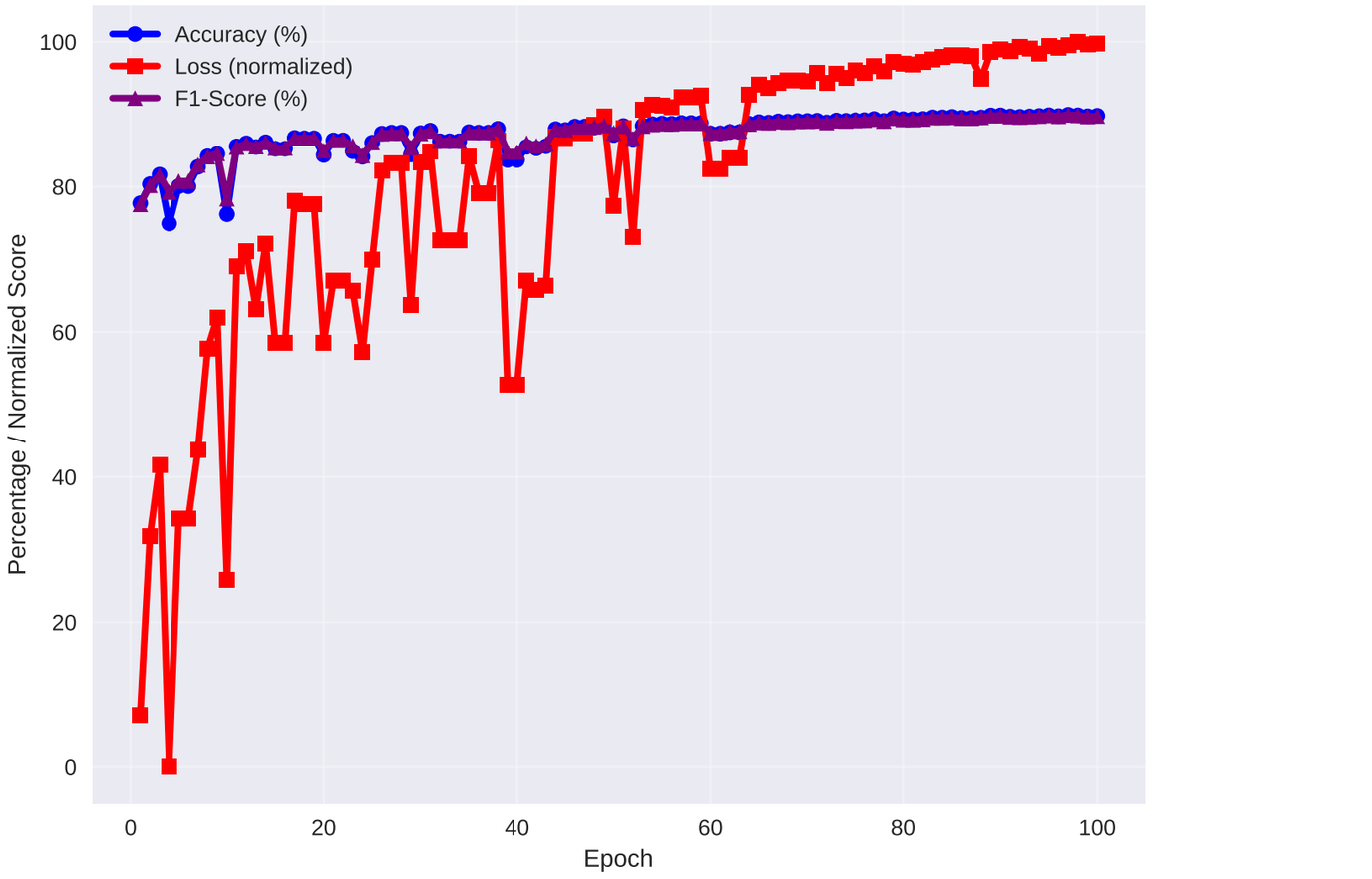
□ Per-Class F1-Score Evolution



□ Inter-Class Variance Evolution



□ Learning Curves Comparison



Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-10 14:35:37
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 89.80%
• Loss: 2.8572
• Average Precision: 0.897
• Average Recall: 0.898
• Average F1-Score: 0.898

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +12.09%
• Riduzione Loss: +3.5174
• Stabilità Accuracy (CV): 0.0341

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.991)
• Peggiore Precision: Classe 6 (0.760)
• Migliore Recall: Classe 8 (0.978)
• Peggiore Recall: Classe 6 (0.667)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.835, R=0.873, F1=0.854
• Classe 1: P=0.991, R=0.968, F1=0.979
• Classe 2: P=0.817, R=0.854, F1=0.835
• Classe 3: P=0.882, R=0.905, F1=0.893
• Classe 4: P=0.822, R=0.838, F1=0.830
• Classe 5: P=0.985, R=0.970, F1=0.977
• Classe 6: P=0.760, R=0.667, F1=0.710
• Classe 7: P=0.938, R=0.971, F1=0.954
• Classe 8: P=0.973, R=0.978, F1=0.976
• Classe 9: P=0.972, R=0.956, F1=0.964

VALUTAZIONE QUALITATIVA:
• Performance Globale: MEDIA
• Stabilità Training: MODERATAMENTE STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
```